Here are two base-36 numbers.

MZY18A
AQ1

What is their product as a base-36 number? Convert this base-36 product to base-10.

Let's go step by step.

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1. Convert each base-36 number to base-10

Recall:

Digits 0–9 = 0–9, A = 10, B = 11, …, Z = 35.

$MZY18A_{36}$

Digits:

- M = 22
- Z = 35
- Y = 34
- 1 = 1
- 8 = 8
- A = 10

$$\begin{aligned} MZY18A_{36} &= 22 \cdot 36^5 + 35 \cdot 36^4 + 34 \cdot 36^3 \\ &\quad + 1 \cdot 36^2 + 8 \cdot 36 + 10 \\ &= 1{,}390{,}630{,}330 \end{aligned}$$

$AQ1_{36}$

Digits:

- A = 10

- Q = 26
- 1 = 1

$$AQ1_{36} = 10 \cdot 36^2 + 26 \cdot 36 + 1 = 13{,}897$$

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2. Multiply (in base-10)

$$1{,}390{,}630{,}330 \times 13{,}897 = 19{,}325{,}589{,}696{,}010$$

So the **base-10 product** is:

$$\boxed{19{,}325{,}589{,}696{,}010}$$

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3. Convert the product back to base-36

Converting $19{,}325{,}589{,}696{,}010$ to base-36 gives the digits:

$$[6,\ 30,\ 22,\ 1,\ 33,\ 4,\ 28,\ 16,\ 10]$$

Mapping to base-36 symbols:

- 6 → 6
- 30 → U
- 22 → M
- 1 → 1
- 33 → X
- 4 → 4
- 28 → S
- 16 → G
- 10 → A

So the **base-36 product** is:

$$\boxed{6UM1X4SGA_{36}}$$

✅ Final Answers

- **Product in base-36: 6UM1X4SGA**
- **Product in base-10: 19,325,589,696,010**

If you want, I can also show the multiplication *directly in base-36* or write a quick script-style method for doing these conversions fast.